



Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY BACHELOR OF TECHNOLOGY

Image Processing (303105381)

7th SEMESTER

Parul Institute of Engineering Technology

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
DATA SCIENCE

Lab Manual

Academic Year : 2026 - 2027

CERTIFICATE

This is to certify that

Jaiswar Abhishek with enrolment no. **2303031240448** has successfully completed her laboratory experiments in the subject (subject code) of **Image Processing laboratory (303105381)** from the department of CSE in **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** during the academic year 2026 – 2027 .



Date of Submission: Staff In Charge:

Head of Department:

INDEX

S.No	Practical Name	From	To	Date of Start	Date of Submission	Sign	Marks
1.	To study and implement basic commands of MATLAB required for digital image processing technique, and various image file formats.						
2.	To study the effect of down sampling and quantization techniques on the grayscale image.						
3.	Write MATLAB code to enhance the visual quality of the image using point processing techniques, namely a) Image negative, b) Logarithmic transformation, and c) Power law transformation and gray level slicing technique.						
4.	Write MATLAB code to display the histogram of the image and enhance the visual quality of the image with the help of histogram equalization technique.						
5.	Write MATLAB code to perform the bit-plane slicing method on grayscale image.						
6.	Write MATLAB code to demonstrate that the convolution in spatial domain is equivalent to multiplication in the frequency domain.						
7.	Write MATLAB code to restore grayscale image from noisy image with the help of image restoration techniques.						
8.	Write MATLAB code to perform Min, Median & Max Filtering on grayscale image.						
9.	Write MATLAB code to blur the image using Ideal low pass, Butterworth low pass and Gaussian Low pass filter.						
10.	Write MATLAB code to blur the image using Ideal High pass, Butterworth High pass and Gaussian High pass filter.						

Practical – 01

Aim: To study and implement basic commands of MATLAB required for digital image processing techniques, and various image file formats.

Software Required:

- MATLAB
- Image Processing Toolbox

Commands:

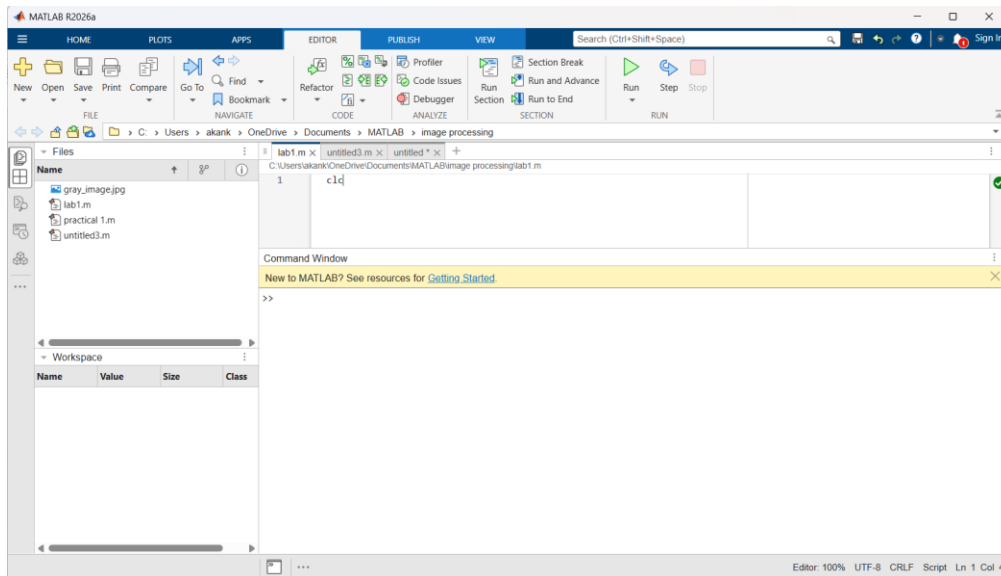
1. `clc` –

- Clears all text from the Command Window.
- Helps keep the MATLAB workspace output clean.

Syntax:

`clc`

Output:



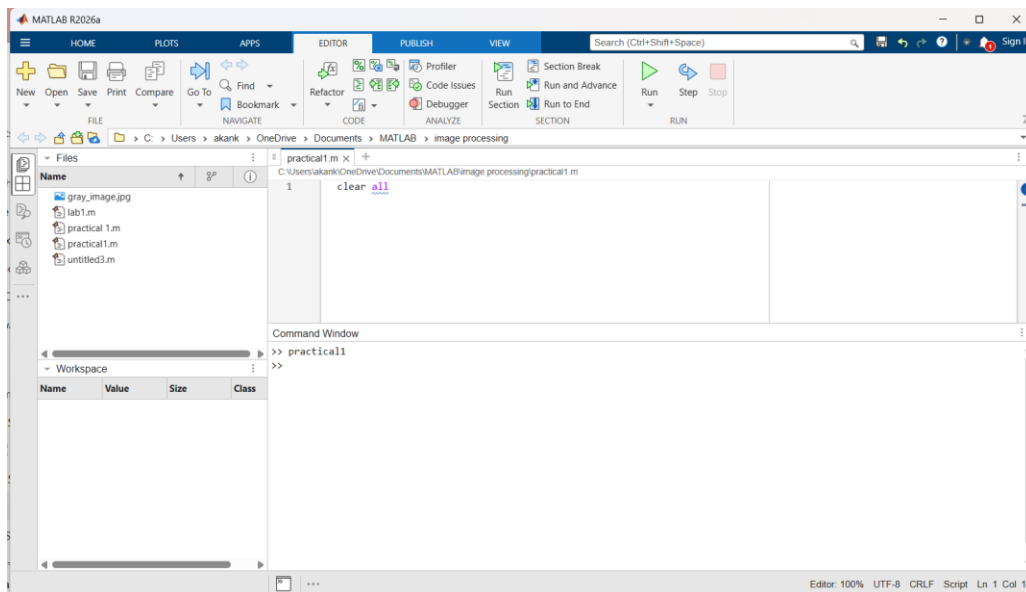
2. `clear all` –

- Removes all variables from the workspace.
- Used to start the program with a fresh workspace.

Syntax:

`clear all`

Output:



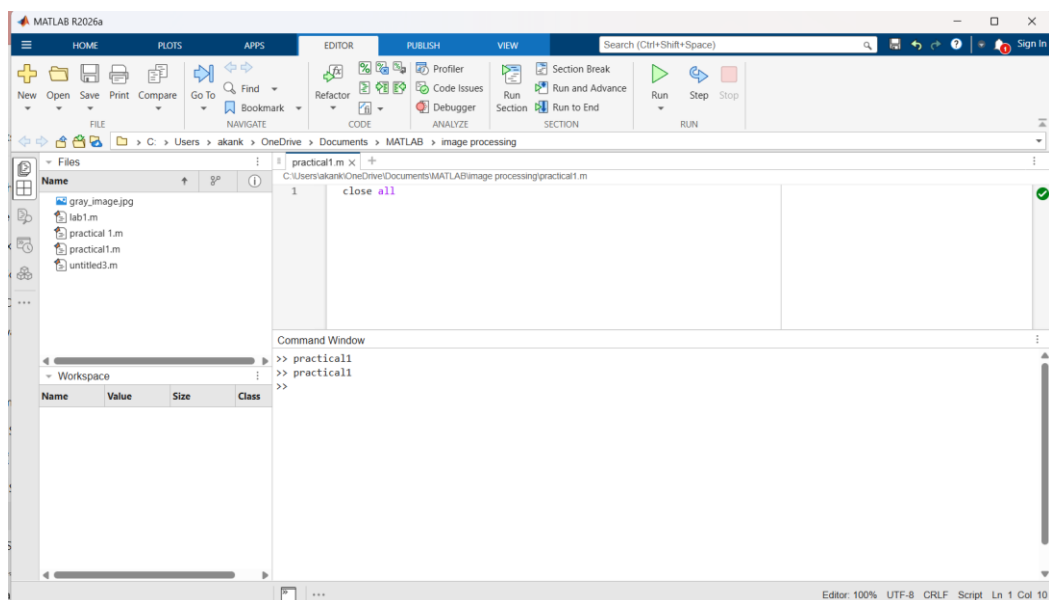
3. close all –

- Closes all open figure windows.
- Useful before displaying new images or plots.

Syntax:

close all

Output:



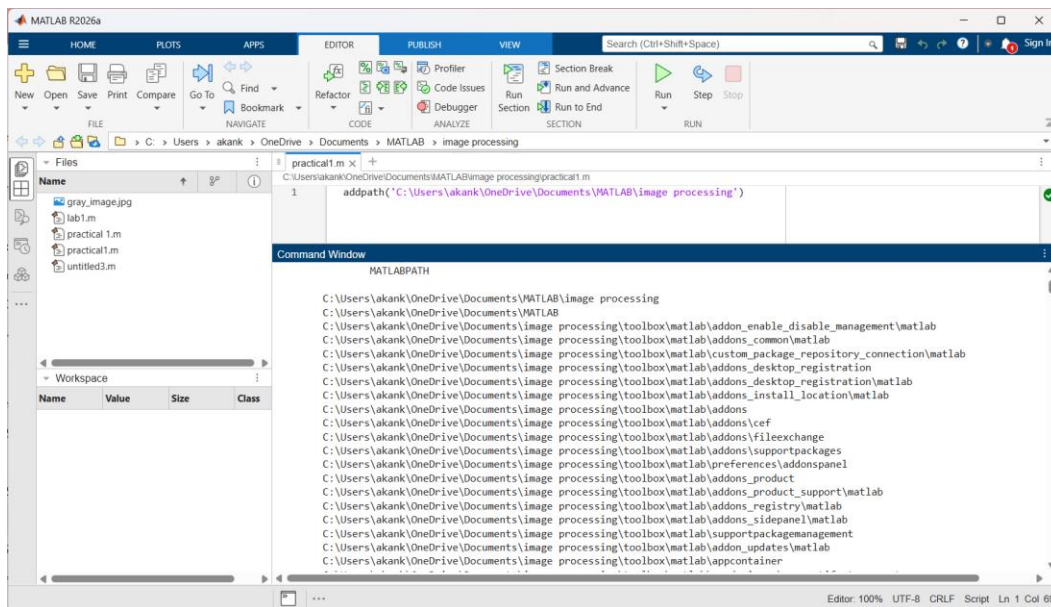
4. addpath –

- Adds a folder to MATLAB's search path.
- Allows MATLAB to access files from that folder.

Syntax:

`addpath('folder_path')`

Output:



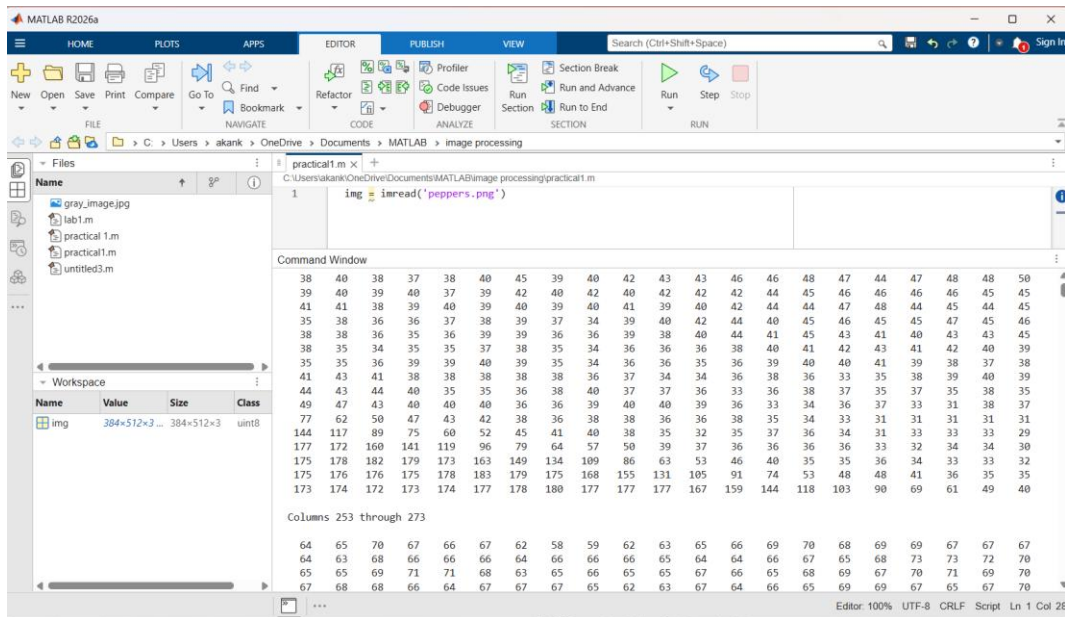
5. imread –

- Reads an image from a file and stores it as a matrix.
- Supports formats like JPG, PNG, BMP, etc.

Syntax:

`img = imread('image.jpg');`

Output:



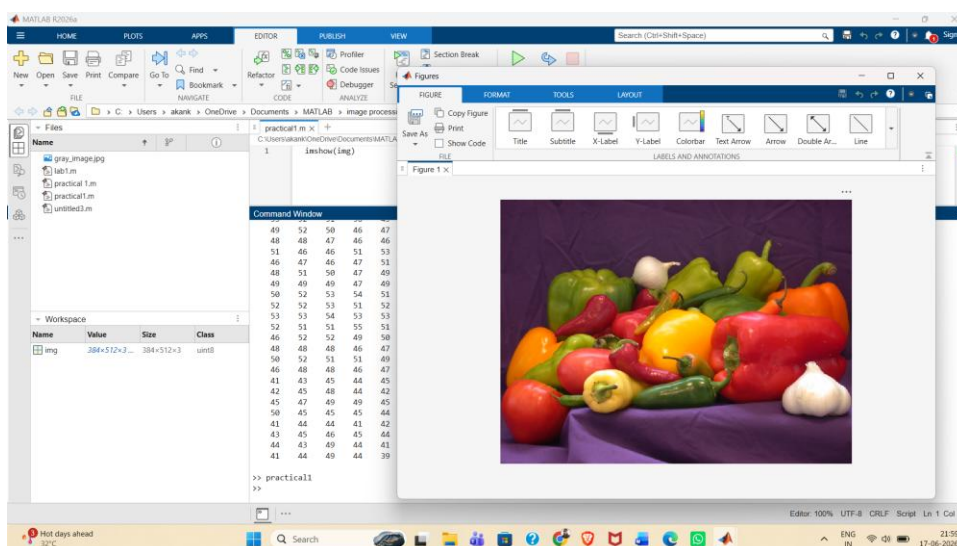
6. imshow –

- Displays an image in a figure window.
- Used for visualizing images in MATLAB.

Syntax:

imshow(img)

Output:



7. title –

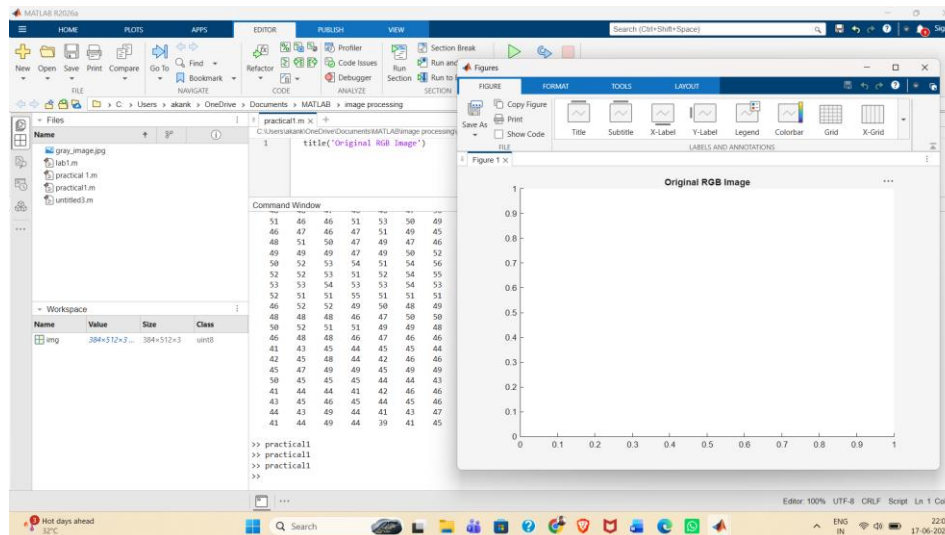
- Adds a title to an image or graph.

- Helps identify the displayed output.

Syntax:

title('Image Title')

Output:



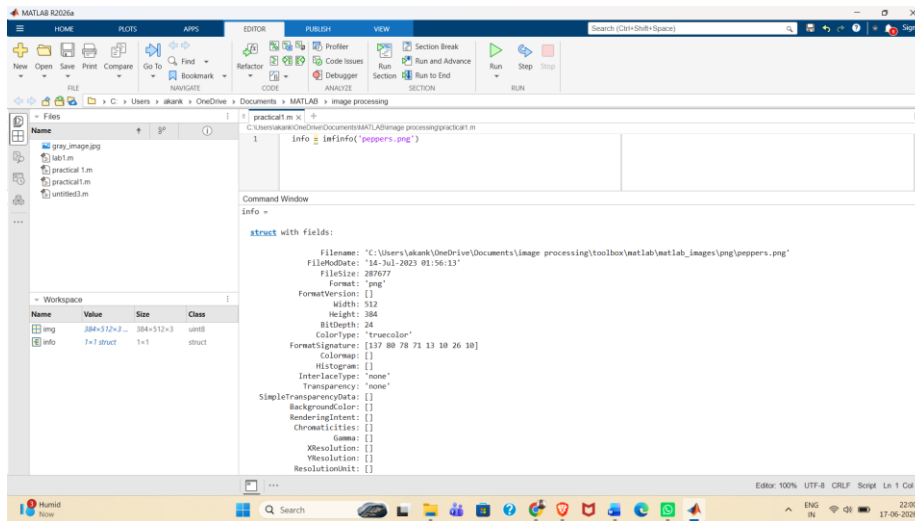
8. imfinfo –

- Displays information about an image file.
- Provides details such as size, format, and bit depth.

Syntax:

info = imfinfo('image.jpg')

Output:



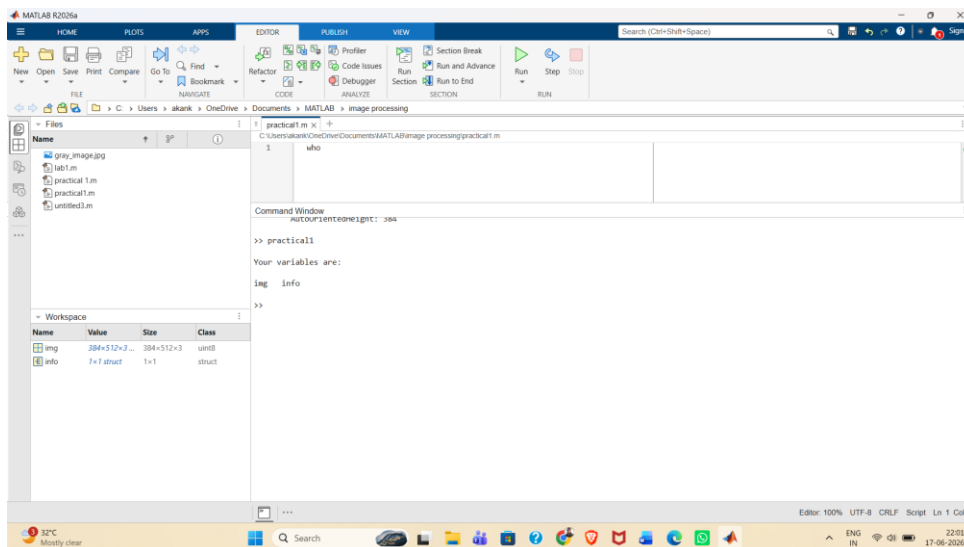
9. who –

- Lists all variable names present in the workspace.
- Used to check currently available variables.

Syntax:

who

Output:



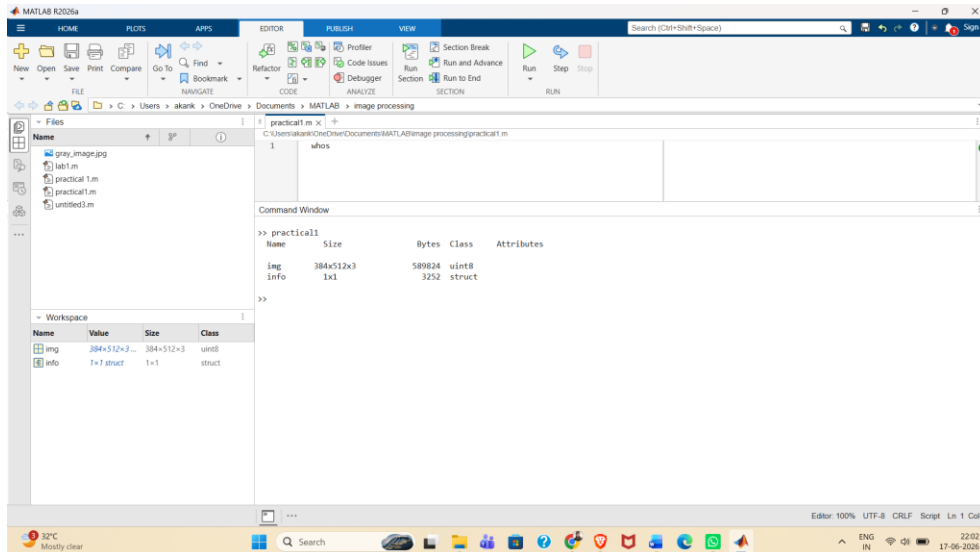
10. whos –

- Displays detailed information about workspace variables.
- Shows size, datatype, and memory usage of variables.

Syntax:

whos

Output:



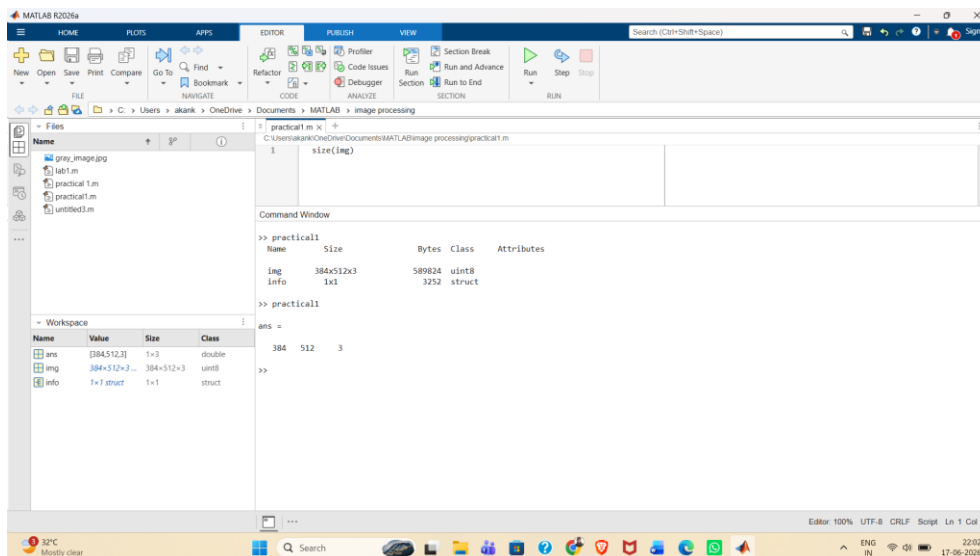
11. size –

- Returns the dimensions of a matrix or image.
- Useful for finding the number of rows, columns, and channels.

Syntax:

size(img)

Output:



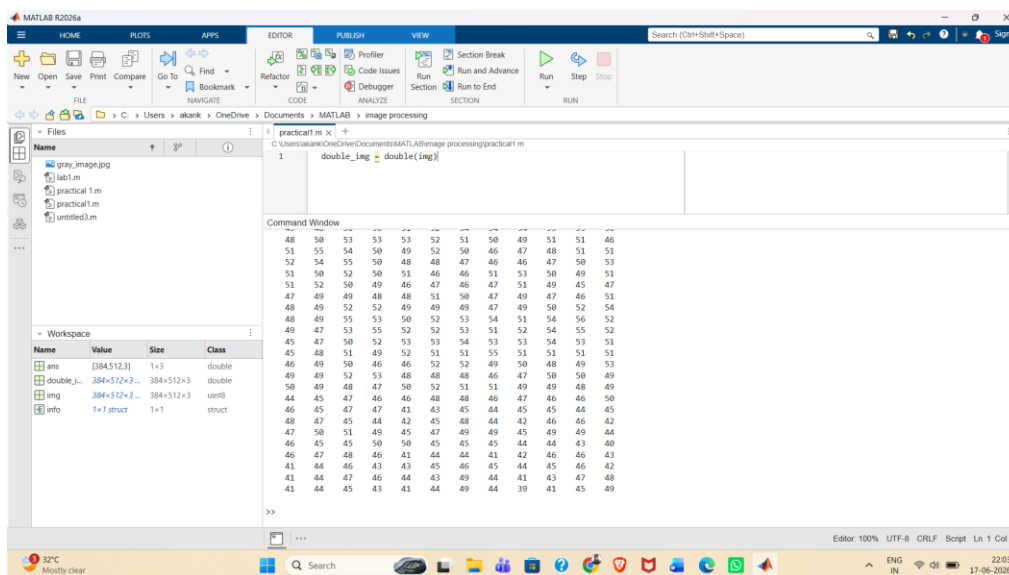
12. double –

- Converts data into double-precision format.
- Required for performing mathematical operations on images.

Syntax:

`new_image = double(old_image)`

Output:



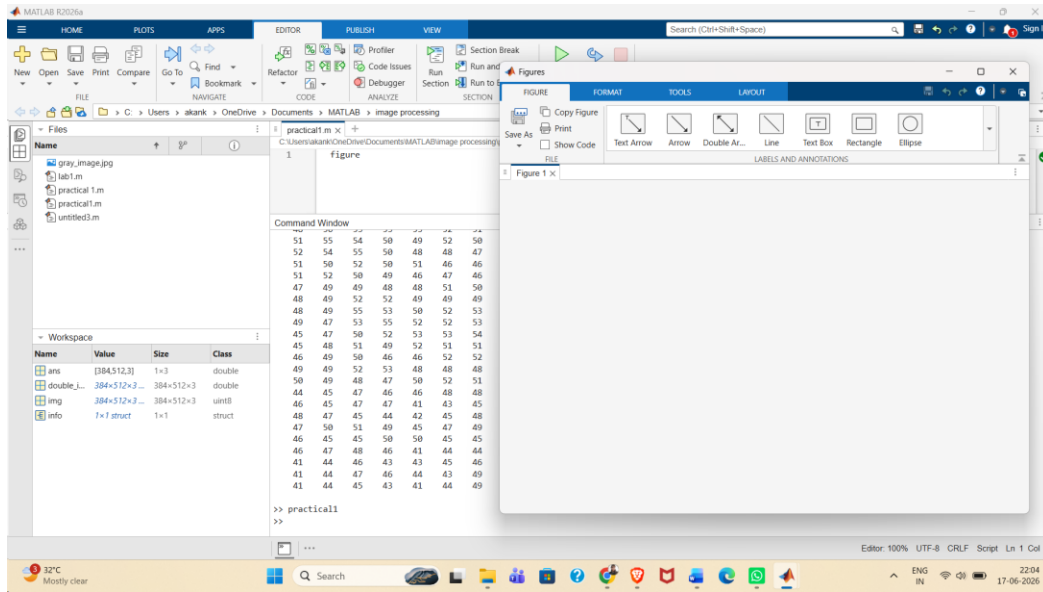
13. figure –

- Creates a new figure window.
- Used to display images and plots separately.

Syntax:

`figure`

Output:



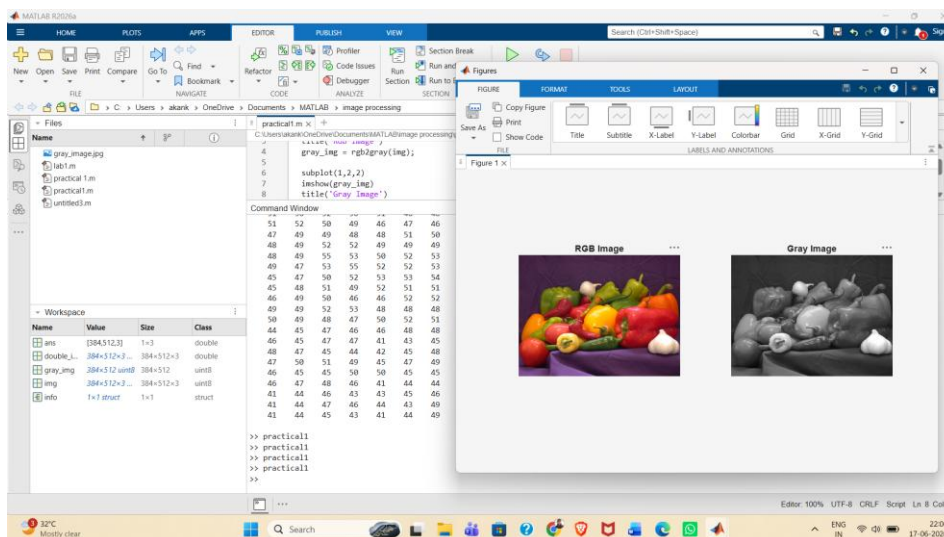
14. subplot –

- Divides a figure window into multiple sections.
- Used to display multiple images or plots in a single figure.

Syntax:

subplot(rows, columns, position)

Output:



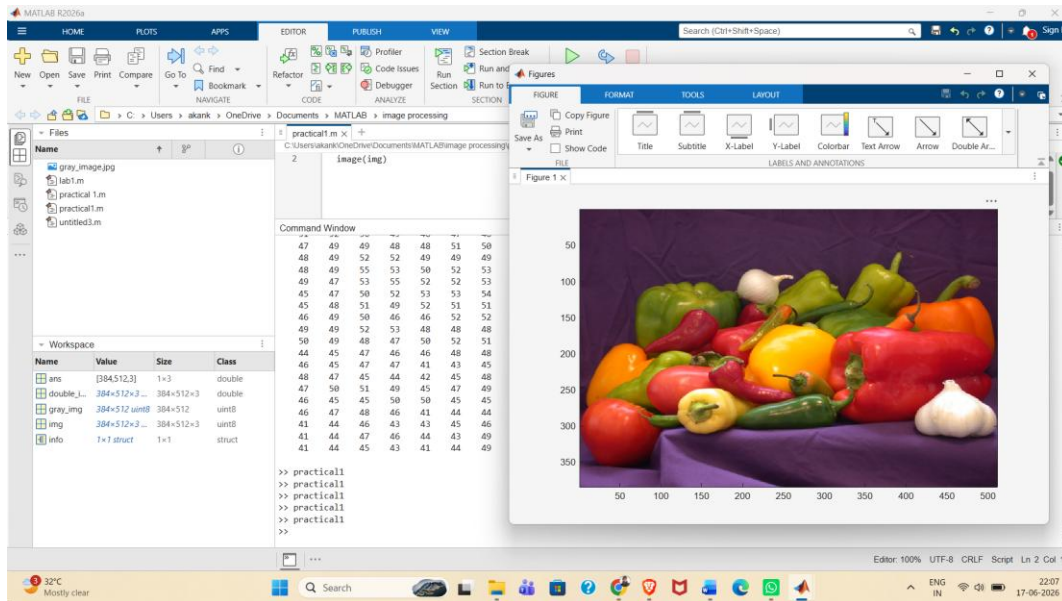
15. image –

- Displays image data stored in a matrix.
- Used to visualize image matrices.

Syntax:

`image(image_name)`

Output:



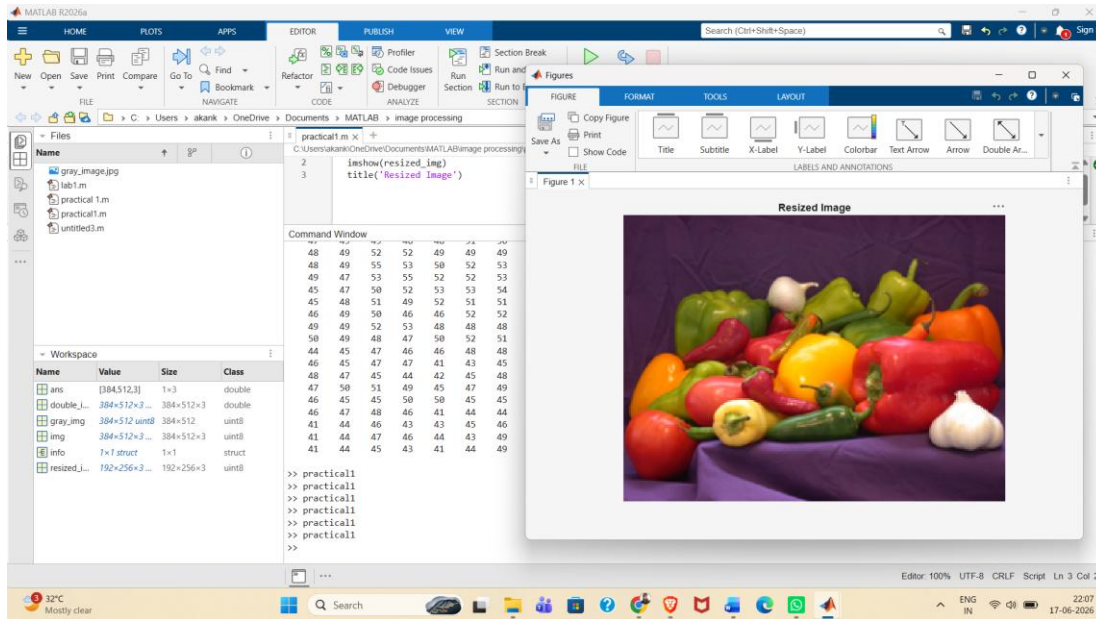
16. imresize –

- Changes the size of an image.
- Used to enlarge or reduce image dimensions.

Syntax:

`resized_image = imresize(original_image, scale)`

Output:



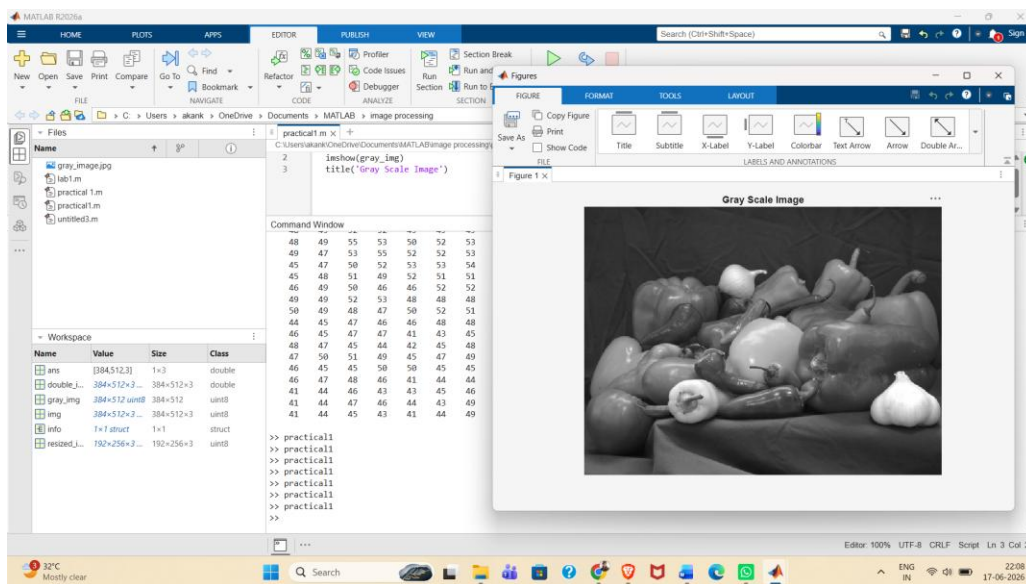
17. RGB to Gray Scale Conversion (rgb2gray) –

- Converts a color (RGB) image into a grayscale image.
- Reduces the image from three color channels to a single intensity channel.

Syntax:

`gray_image = rgb2gray(rgb_image)`

Output:



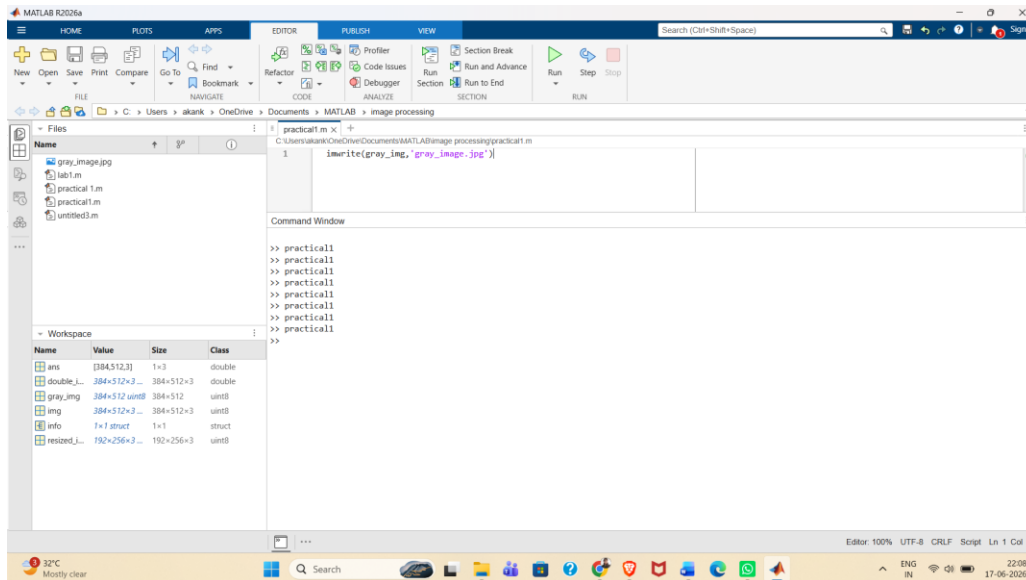
18. imwrite –

- Saves an image to a specified file format.
- Used to store processed images on disk.

Syntax:

```
imwrite(image, 'filename.extension')
```

Output:



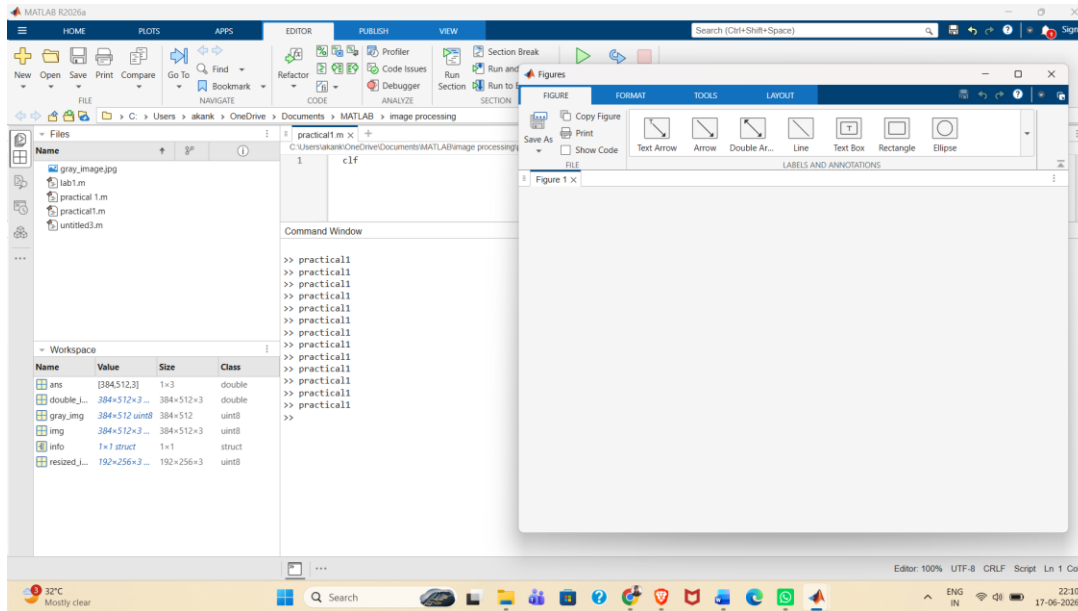
19. clf –

- Clears the contents of the current figure window.
- Does not close the figure window.

Syntax:

clf

Output:



Conclusion:

Thus, the basic MATLAB commands used in image processing were studied and implemented successfully. Different image file formats were analyzed, and operations such as image reading, displaying, resizing, saving, and obtaining image information were performed using MATLAB commands.